INCH-POUND
MIL-DTL-8878G
30 October 1999
Superseding
MIL-DTL-8878F
22 FEBRUARY 1999

DETAIL SPECIFICATION

TURNBUCKLES, POSITIVE SAFETYING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers positive safetying turnbuckle component parts.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

* **Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Richmond, ATTN: DSCR-VBD, Richmond, VA 23297-5610 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.**

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SPECIFICATIONS

FEDERAL

QQ-A-225/6	**-**	**Aluminum Alloy 2024 Bar, Rod, and Wire; Rolled, Drawn or Cold Finished.**
QQ-P-416	**-**	**Plating, Cadmium (Electrodeposited).**
WW-T-700/3	**-**	**Tube, Aluminum Alloy, Drawn, Seamless, 2024.**

DEPARTMENT OF DEFENSE

MIL-S-6050	**-**	**Steel, Chrome-Nickel-Molybdenum (8630) Bars and Reforging Stock (Aircraft Quality).**
MIL-H-6088	**-**	**Heat Treatment of Aluminum Alloys.**
MIL-T-6736	**-**	**Tubing, Chrome-Molybdenum (4130 or 8630) Steel, Seamless and Welded.**
* **MIL-S-6758**	**-**	**Steel, Chrome-Molybdenum (4130) Bars and Reforging Stock (Aircraft Quality).**
MIL-H-6875	**-**	**Heat Treatment of Steel, Process for.**
MIL-A-8625	**-**	**Anodic Coating, For Aluminum and Aluminum Alloys.**
MIL-C-16173	**-**	**Corrosion Preventive Compound, Solvent Cutback, Cold-Application.**

STANDARDS

FEDERAL

FED-STD-H28/20	**-**	**Screw – Thread Standards for Federal Services Section 28 Inspection Methods for Acceptability of UN, UNR, UNJ, M, and MJ Screw Threads.**

DEPARTMENT OF DEFENSE

MIL-STD-130	**-**	**Identification Marking of U.S. Military Property.**
MS21251	**-**	**Turnbuckle Body, Clip Locking.**
MS21252	**-**	**Clevis End, Turnbuckle, Clip Locking.**
MS21253	**-**	**Clevis End, Turnbuckle, Clip Locking (for Bearing).**
MS21254	**-**	**Eye End, Turnbuckle, Clip Locking.**
MS21255	**-**	**Eye End, Turnbuckle, Clip Locking.**
MS21256	**-**	**Clip, Locking, Turnbuckle.**
MS33736	**-**	**Turnbuckle Assemblies, Clip Locking of.**
* **MS21260**	**-**	**Stud End, Turnbuckle, Clip Locking.**

(Unless Otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4, Philadelphia, PA 19111-5094.)

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2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASQC Z1.4 -Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-E8 - Materials, Metallic, Tension Testing Of.
ASTM-B21 - Standard Specification for Naval Brass Rod, Bar and Shapes E2 (DoD adopted).
ASTM-E140 - Metals, Standard Harness Conversion Tables For
ASTM-A313 - Chromium-Nickel Stainless and Heat-Resisting Steel Spring Wire, Standard Specification For
ASTM-A580 - Standard Specification for Stainless and Heat-Resisting Steel Wire (Not Current Document).
ASTM-A630 - Standard Test Methods for Determination of Tin Coating Weights for Hot-Dip and Electrolytic Tin Plate.
ASTM-A967 - Chemical Passivation Treatments for Stainless Steel Parts
ASTM-A564 - Hot-Rolled and Cold Finished Age-hardening Stainless Steel Bars and Shapes
*** ASTM-B135 - Seamless Brass Tube.**

*** (Application for copies should be addressed to the American Society for Testing and Materials, Inc., 100 Barr Harbor Drive, Conshocken, PA 19428-2959.)**

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

AMS 2431 -- Peening Media General Requirements.
AMS 5643 - Steel, Corrosion Resistant, Bars, Wire, Forgings, Tubing, and Rings 16Cr – 4.0Ni – 0.30 (Cb+Ta) – 4.0Cu Solution Heat Treated, Precipitation Hardenable.
*** AMS 6360 - Steel Tubing, Seamless, 0.95Cr - 0.20Mo. (0.28 – 0.33C)**

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive Warrendale PA 15096-0001; telephone (412)772-1616.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 MS standards. The individual item requirements shall be as specified herein and in accordance with the applicable MS sheet. In the event of any conflict between requirements of this specification and the MS sheet, the latter shall govern.

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3.2 Qualification. The turnbuckle component parts furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.3 Material. The material used in the manufacture of turnbuckle parts and locking clips shall be as specified herein.

3.3.1 Turnbuckle bodies. The turnbuckle bodies shall be made of one of the following materials.

3.3.1.1 Brass bodies. Brass turnbuckle bodies shall be made from UNS C48200 or UNS C46400,
*** hard tempered (H04) per ASTM-B21 or UNS C33000 seamless brass tubing per ASTM-B135. The 15 percent minimum elongation value in Table II of ASTM-B135 is changed to 9 percent minimum. Marking by impression stamping, if used, shall be done before stress relieving. Brass turnbuckle bodies shall be stress relieved at 450 degrees F for two hours after fabrication.**

3.3.1.1.1 Hardness. Finished brass turnbuckle bodies shall have a minimum hardness of R15T-70.
*****

3.3.1.2 Steel bodies. Steel turnbuckle bodies shall be made from steel bar per MIL-S-6050 or MIL-S-6758 or seamless steel tubing per AMS-6360, except that the tensile strength shall be 75,000 PSI minimum.

3.3.1.3 Aluminum bodies. Aluminum turnbuckle bodies shall be made from aluminum alloy temper T4 or T851 per QQ-A-225/6 or tubing material per WW-T-700/3, except that heat treatment shall be T81 or T851 per MIL-H-6088.

3.3.2 Clevis and eye turnbuckle ends. Clevis and eye turnbuckle ends shall be forged or machined from 17-4PH per AMS 5643 or ASTM-A564 in the H1150 or H1150M condition or cold-rolled or cold-drawn carbon or alloy steel, per MIL-S-6578 of which the sulfur content does not exceed .050 percent, and the phosphorus content does not exceed .040 percent, provided the finished parts conform to applicable requirements of this specification. The carbon or alloy steel is inactive for new designs and the 17-4PH material should be used in current applications wherever possible. Ends not fabricated from heat treated material shall be heat treated per MIL-H-6875 to conform to this specification and the applicable MS sheet.

*** 3.3.3 Locking clip. The locking clip shall be made of corrosion-resisting steel composition UNS – S30200/S30400, condition B, cold finished per ASTM-A580 or ASTM-A313.**

3.4 Construction. The construction of turnbuckle components shall be as specified herein and on MS21251, MS21252, MS21253, MS21254, MS21255 and MS21256.

3.4.1 Turnbuckle body and ends. Turnbuckle body and ends shall contain facilities for employing a locking clip which will provide positive safetying of mating parts in assembly.

3.4.1.1 Positive safetying. Positive safetying, for purpose of this specification, is defined as preventing rotational movement between turnbuckle body and end within the requirements of this specification without structural failure of the components or locking clip.

3.4.2 Locking clip. The locking clip shall be constructed as specified herein and on MS21256. It shall be capable of being installed manually without the use of tools and shall be self-retaining in assembly.

3.5 Protective treatment. Turnbuckle parts shall be treated for corrosion protection as specified.

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3.5.1 Steel parts. Carbon and alloy steel turnbuckle parts shall be cadmium plated per type II, class 2 of QQ-P-416. Parts made of 17-4PH shall be (a) passivated after the removal of all oil or grease by submerging into a solution containing four parts water to one part commercial nitric acid, specific gravity 1.389 to 1.408, at 125 to 135 degrees F, for 20 to 40 minutes (equivalent method of ASTM-A967 may be used) or (b) glass bead blasted using beads conforming to AMS 2431. Unless otherwise specified, either method for 17-4PH is acceptable. The bore of steel turnbuckle bodies shall be coated with corrosion preventive compound conforming to grade 1 of MIL-C-16173.

3.5.2 Aluminum parts. Aluminum alloy shall be anodized per MIL-A-8625, Type II, Class 1.

3.6 Dimensions. The dimensions of turnbuckle bodies, turnbuckle ends, and the locking clip shall conform to those specified on the applicable MS sheet.

3.6.1 Threads. The internal threads in the turnbuckle bodies shall conform to FED-STD-H28/20 UNF-3B. Concentricity between tap hole and threads shall be such that the locking clips can be installed with the terminals engaged to the cross hole for all locking slots (see 4.6.5.1).

3.7 Physical properties.

3.7.1 Rotational movement. The total angular clockwise and counterclockwise rotational movement between turnbuckle body and turnbuckle end with locking clip installed shall not exceed that specified in table I when subjected to 10 percent of the specified torsion load (see 4.6.1).

TABLE I. Torsion test loads and rotational movement.

MS Standard Dash Number	Minimum Torque (in-lbs)	10% of Torsion Load (in-lbs)	Rotation Movement (maximum degrees)
2	8	.8	15
3	26	2.6	11
4	82	8.2	8
5	82	8.2	8
6	163	16.3	7
7	318	31.8	6
8	318	31.8	6
9	523	52.3	5
10	750	75.0	4

3.7.2 Torsion. Turnbuckle bodies and ends shall be capable of withstanding the torsional loads specified in table I, applied clockwise and counterclockwise direction without permanent deformation when tested per 4.6.2.

3.7.3 Tensile fatigue. Turnbuckle bodies and ends shall be capable of withstanding a load equal to 60 percent of the minimum breaking strength (MBS) specified on the applicable MS sheet, applied repeatedly in tension at a rate of approximately one cycle per second for 300 applications without failure, deformation, or permanent increase in overall length of any component part in excess of .001 inch per inch in the direction of the application of load (see 4.6.3).

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3.7.4 Breaking strength. The MBS strength of turnbuckle parts shall not be less than the values specified on the applicable MS sheet (see 4.6.4).

3.7.5 Bending. Turnbuckle ends shall withstand bending through an angle of 90 degrees without cracking or fracture. Locking clip formed radii shall withstand straightening so that the test sample becomes approximately a straight length of wire (see 4.6.6).

3.7.6 Locking clip retention. Locking clips when installed in the turnbuckle assembly shall withstand a removal or disengagement force of 15 lbs when tested per 4.6.5.

3.8 Identification. Turnbuckle bodies and ends shall be marked with manufacturer's name or trademark, and the applicable MS part number per MIL-STD-130. All turnbuckle bodies shall be dated showing the year of manufacture by any permanent means available in MIL-STD-130. Location of the marking shall be as indicated on the applicable MS sheet. Where space will not permit all marking to be located in a single area, the information shall be divided appropriately, and conspicuously located in two adjacent areas. The clip shall be coated with light blue Dykem code DMF (staining color shall be obtained from Dykem Company, 8501 Delport Drive, St. Louis, MO 63114), or equal. Incomplete coverage shall not be cause for rejection. (Equivalent processes shall be evaluated, when requested, through the QPL engineering office.)

3.9 Workmanship. Turnbuckle parts shall be 125 RMS or better and free from burrs, sharp edges, cracks, flaws, pronounced tool marks, or other discontinuities which would prevent proper functioning or performance. Slight burrs in the thread locking area that do not prevent gauging or interfere with installation of the terminal and locking clip are acceptable. Discoloration resulting from heat treatment of 17.4PH will be removed.

3.10 Penetrant inspection. Brass and aluminum turnbuckle bodies shall be fluorescent penetrant inspected per MIL-STD-6866, Type I. Any crack visible with the naked eye shall be cause for rejection.

3.11 Reclaimed materials. Use of reclaimed material is to be promoted to the greatest extent possible.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

- **a. Qualification inspection (see 4.3).**
- **b. Quality conformance inspection (see 4.4).**

4.2 Qualification test facilities. Due to the critical nature of these products and the urgent and ongoing requirement for quality control, qualification test facilities must be readily accessible to government personnel.

4.3 Qualification inspection.

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4.3.1 Sampling instructions. Qualification test samples shall consist of four turnbuckle assemblies containing the component part of the type and size upon which qualification is desired. The four turnbuckle assemblies are to be distributed as specified in table II.

TABLE II. Qualification test sample distribution.

Test	Quantity
Rotational Movement	1
Torsion	Use rotational movement sample
Hardness	Use rotational movement sample
Tensile Fatigue	1
Breaking Strength	1
Locking Clip Retention	1
Bending	Use breaking strength sample

4.3.2 Tests. The qualification tests of turnbuckle component parts shall consist of all the tests specified under 4.5 and 4.6.

4.3.3 Retention of qualifications. To retain qualification, the manufacturer must be able to demonstrate that the company still has the capabilities and facilities necessary to produce the items. The qualifying activity will request appropriate documentation demonstrating this capability at the time qualification retention is required (at least 2 year intervals).

4.4 Quality conformance inspection. Quality conformance tests shall consist of examinations and tests specified under 4.5, 4.6.4, 4.6.5, 4.6.6 and 4.6.7.

*

4.4.1 Inspection lot. An inspection lot shall consist of turnbuckle component parts of the same type, size and material produced under the same manufacturing conditions. The unit of inspection shall be one turnbuckle component part except for hardness and examination of product.

4.4.2 Hardness Sampling. The unit of inspection shall be one brass turnbuckle body per stress relief batch.

4.5 Examinations.

4.5.1 Examination of product. Each sample part shall be examined for conformance to the material, design, dimensional, protective treatment, marking, workmanship and penetrant inspection requirements of this specification and the applicable MS sheet.

*

4.5.2 Sampling. ANSI/ASQC Z1.4 shall be used as a guide in the development of contractor's statistical techniques to assure the components meet all requirements specified herein.

4.6 Tests.

4.6.1 Rotational movement. A turnbuckle assembly containing the sample component part with a locking clip installed, shall be subjected to 10 percent of the applicable torque load specified in table I, applied in a clockwise and counterclockwise direction. The total relative angular rotation between the turnbuckle end and turnbuckle body (less the free play) from the extreme of one direction (clockwise) to the extreme of the other (counterclockwise) shall be measured for conformance to 3.7.1.

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4.6.2 Torsion. A turnbuckle assembly containing the sample component part with a locking clip installed shall be subjected to the applicable torque load specified in table I, applied in a clockwise and counterclockwise direction for five complete cycles. The turnbuckle shall then be disassembled and the sample shall be inspected for evidence of failure per 3.7.2. There shall be no evidence of cracks in the sample turnbuckle part of locking clip. Deformation of the turnbuckle or locking clip is acceptable if parts can be disassembled by hand.

4.6.3 Tensile fatigue. A turnbuckle assembly containing the sample component part shall be subjected to a tension load equal to 60 percent of the MBS specified on the applicable MS sheet for the sample, applied repeatedly in tension at a rate of approximately one cycle per second for 300 applications. The turnbuckle assembly shall be disassembled and the sample parts shall be inspected for conformance to 3.7.3. Elongation of the sample turnbuckle part(s) shall be determined per ASTM-E8.

4.6.4 Breaking strength. A turnbuckle assembly containing the sample component part shall be tested for MBS by an appropriate means per ASTM-E8. The MBS value is that specified on the applicable MS sheet. There shall be no evidence of failure in the sample part. It may be necessary to fabricate special high strength fixtures for attaching to the parts to be tested in order to develop the required loads.

4.6.5 Locking clip retention. A locking clip shall be installed in a turnbuckle per the instructions specified on MS33736. A direct pull of 15 lbs shall be applied to the locking clip as shown on figure 1. The locking clip shall not break nor shall the retention feature become disengaged.

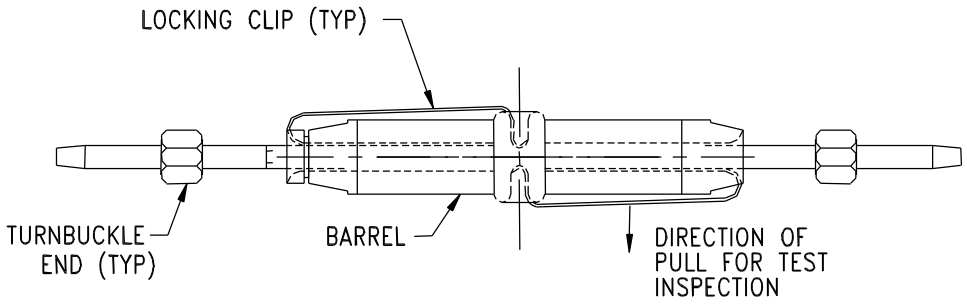


FIGURE 1. Locking clip retention.

4.6.5.1 Functional acceptability. The turnbuckle body and mating terminals must be assembled and the locking clip inserted in each end of the turnbuckle body. After removing clips, the terminals are rotated one half turn and the locking clip inserted once again. This functional assembly requirement tests every combination of the locking capability of the assembly.

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4.6.6 Bending. Turnbuckle parts shall be tested for bending as follows:

4.6.6.1 Turnbuckle ends. Sample turnbuckle ends that pass the breaking strength test (4.6.4) shall be held in a square-nosed vise and the shank bent through an angle of 90 degrees. There shall be no evidence of failure or cracking when inspected.

4.6.6.2 Locking clip. Locking clip must withstand straightening with the use of pliers. The various formed radii shall be straightened so that the sample becomes approximately a straight length of wire. This is to insure that brittleness has not been induced in the formed areas of the clip. There shall be no evidence of failure (break) or cracking.

4.6.7 Hardness. Hardness testing shall be performed on the outer surface of the brass turnbuckle body in accordance with ASTM-E140.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Turnbuckles and component parts are intended for use in the construction of aircraft and aircraft accessories.

6.2 Acquisition requirements. Acquisition documents must specify the following:

- **a. Title, number, and date of this specification.**
- **b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).**
- **c. Name of part.**
- **d. Part number.**
- **e. Packaging requirements (see 5.1).**

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Product List QPL-8878 whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to contact the activity responsible for the Qualified Products List to determine the specific requirements that must be met for qualification. The * **activity responsible for the Qualified Products List is Defense Supply Center Richmond, ATTN: DSCR-VBD, Richmond, VA 23297-5610and information pertaining to qualification of products may be obtained from that activity.**

6.4 Subject term (key word) listing.

Connectors
Fittings
Wire rope assemblies

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTEREST

Custodians:
Army – CR4
Navy - AS
Air Force – 99

*

Review activities:
Air Force – 82

Preparing activity:
DLA - GS

(Project 1560-0265)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-8878G	**2. DOCUMENT DATE (YYYYMMDD)** 19991030

3. DOCUMENT TITLE **TURNBUCKLES, POSITIVE SAFETYING**

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE SUPPLY CENTER RICHMOND	b. TELEPHONE *Include Area Code)* (1) Commercial (804) 279-3875 (2) AUTOVON 695-3875
c. ADDRESS *(Include Zip Code)* DEFENSE SUPPLY CENTER RICHMOND, ATTN: DSCR-VBD, 8000 JEFFERSON DAVIS HIGHWAY, RICHMOND, VA 23297-5610	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533, Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

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